TONIGHT

Groups nobody named, and how to find them

How clustering thinks, thirty shares compressed to three forces, ten thousand complaints read by meaning. Then the finale report.

Data Analysis & Machine Learning for Business

Andrey Shumilin

From points without answers to a report.

01	The quiz and HW1: ideas for HW2	18:10
02	How clustering thinks	Demo 1
03	Thirty shares, three forces	19:40, Demo 2
04	What are ten thousand complaints about?	Demo 3
05	The finale report and a summary	20:40

HW2 is still open: **due October 21, 18:00**.

Four ideas worth a second look.

01

A total needs the mean.

Median 50, mean 100, the shift needs 1,000: **10 workers**, not 20.

02

A join drops rows that have no partner.

139 thousand events with an empty `item_id`. Not sure? A five-row toy table.

03

A straight line can't give a probability.

In session 5 a quarter of the test firms got a value below zero.

04

A p-value is a share of worlds, not the size of the gap.

No evidence of a difference is not no difference.

HW2 due Oct 21, 18:00: **one week left.**

Eight habits for HW2.

01 Trace one case by hand, **in the raw log**.

02 Every finding ends with a decision.

03 One dependence is not the whole story.

04 A recommendation needs its effect, **as a number**.

05 Compare definitions by the answer, not by the count.

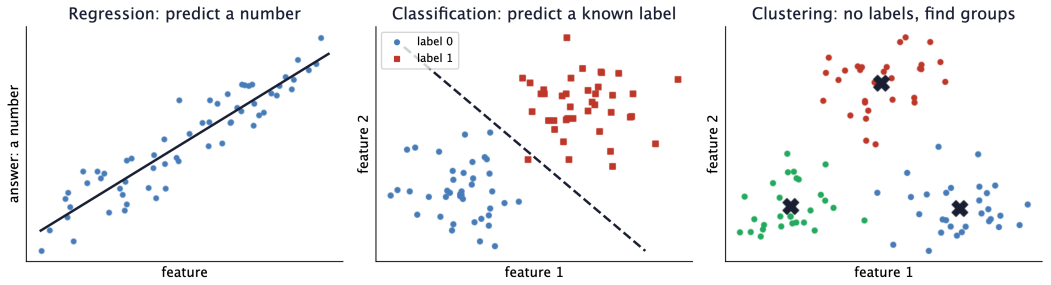
06 **Conclusion first**, then the number, then the action.

07 A chart for a manager **stands on its own**.
Clear to someone outside the context. Every conclusion comes from what is on it.

08 “No effect” or **“no effect the way I looked”**?
Median split: nothing for weight. Heavier than 2 kg or top 5%: a clear effect.

The format worked: **hypotheses before code, verdicts after.**

Clustering has no answer column.



TASK	WHAT COMES OUT	ANSWER COLUMN?	HOW WE JUDGE IT
Regression	a number	yes	average miss, R^2
Classification	a known label	yes	recall, precision, AUC, cost
Clustering	groups nobody named	**no**	can you name them, act on them, get them again?

Ask Where in your work is there data without an answer column?

Six places you will meet it.

01 Customer segments → different offers

02 Ten thousand complaints → what to fix first

03 Branches like this one → fair comparison

04 Shares that move together → diversification

05 A day or a client in no group → anomaly

06 A step before optimisation → **Ozon**



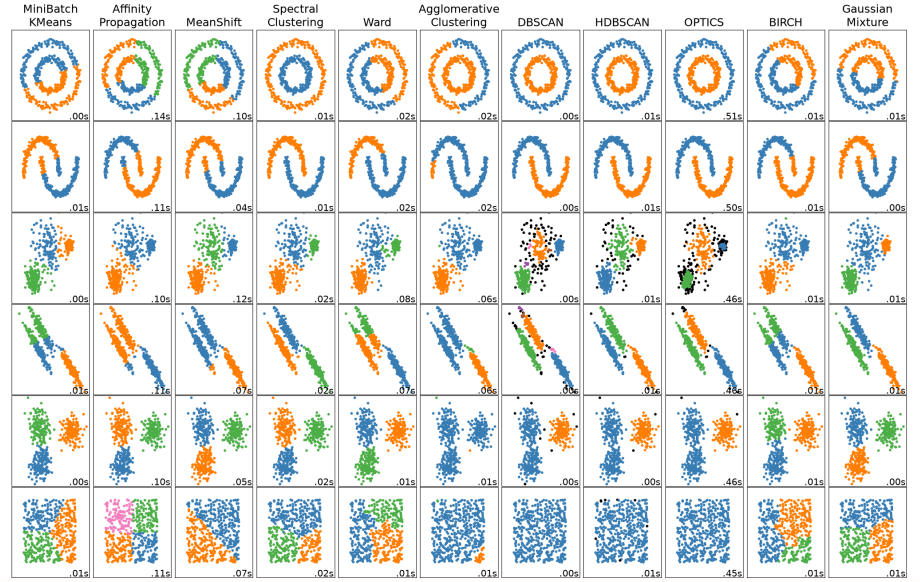
OZON TECH, HABR, 2025

Cabinets clustered by how full they are: **Affinity Propagation, then agglomerative**. Clustering narrows the search before placing goods.

habr.com/ru/companies/ozontech/articles/904044

Maths is not only ML.

Every algorithm has its own idea of a group.



scikit-learn.org › Clustering › Overview of methods

k-means

Round groups around centers. Cuts moons and rings. Draws groups where there are none.

DBSCAN, HDBSCAN

Dense areas, any shape. Loners are left out as noise.

Hierarchical

A tree of merges. Cut it at any level.

Ask Your customer data: which row does it look like?

LIVE CODING, SWITCH CONTEXT

`01-how-clustering-thinks.ipynb`

Demo 01

k-means by hand, then density, then a tree. Settings, and why they matter.

THE RULE OF THE DEMO

k-means always returns k groups.

Even on a cloud that has none.

1	k-means by hand on 8 points
2	One cloud, k = 3: **three slices anyway**
3	Scale first: salary vs years
4	Choosing k: elbow and silhouette
5	Moons: DBSCAN and its radius
6	HDBSCAN: no radius, **use it first**
7	A tree of merges

Ask The Taiwan cloud: how many groups do you see?

Nine ways a clustering misleads.

01 **No right answer.**

02 **It always finds k groups.**

03 You choose k, eps or the smallest group.

04 Units decide the distances.

05 k-means sees only round groups.

06 Outliers drag centers.

07 Another seed, other groups.

08 Many columns blur distances.

09 A cluster can be one template copied.

Ask Which of these would you not notice without a picture?

Can the data find the causes by itself?

WEEKLY RETURNS

195 × 30

weeks × shares



THREE COLUMNS

market

oil

gold

Most of the movement has a few common causes. **That is PCA.**

02-pca-market-factors.ipynb

Demo 02

Thirty shares, three forces. PCA as a rotation of axes.

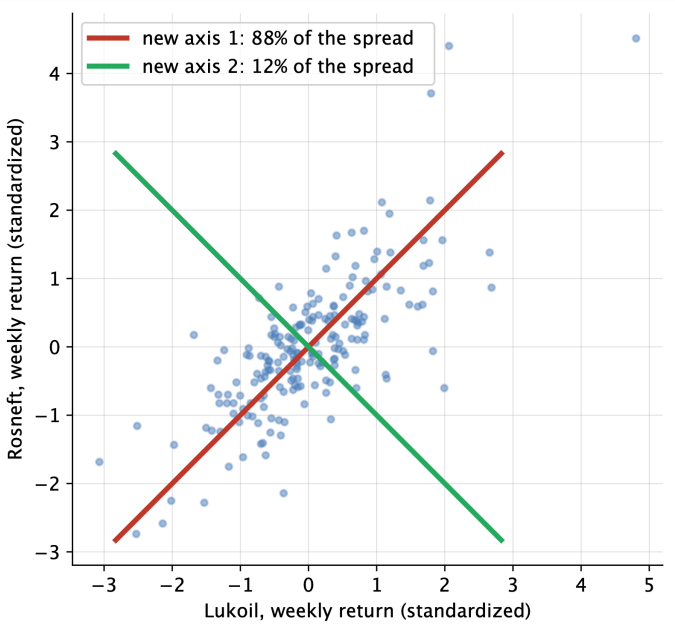
A COMPONENT IS A WEIGHTED SUM

axis 1 = w_1 × Lukoil + w_2 × Rosneft

Rotating is multiply and add.

1	Prices to weekly returns
2	Two oil shares: an oval
3	Rotate the axes, **keep the long one**
4	Thirty shares: one big force
5	Name the forces: market, oil, gold
6	PC1 next to the MOEX index

Same points, new axes.



TWO OIL COMPANIES

Axis 1 runs along the oval. **One number per week instead of two.**

THIRTY SHARES, THREE FORCES

market **oil** **gold**

PCA rebuilt the index **without seeing it.** It can only rotate: straight axes.

Words become numbers. Rare words weigh more.

WORD COUNTS

`0 1 0 0 3 1 0 2 …`

one column per word, how many times it appears

TF-IDF

tf × log(N / df)

tf: times in this text · **N**: all texts · **df**: texts with the word

WORD IN ONE COMPLAINT	TF	DF, OF 10,000	LOG(N / DF)	TF-IDF
the	3	9,800	0.02	**0.06**
payment	2	2,000	1.61	**3.22**
Zelle	1	100	4.61	**4.61**

Frequent everywhere: weight near zero. **Rare and present here: the text is about it.**

An encoder turns a text into a point.

“billed two times”

a complaint, any length

→

`billed` `two` `times`

cut into tokens

→

neural network

pretrained, we only download it

→

`0.03 −0.12 0.41`

…

384 numbers

Same length for every text, short or long.

Reads the text, not only which words are in it.

Runs on a laptop, in seconds, for free.

Every text becomes a point. **Points can be clustered.**

Trained to put the same meaning nearby.

TRAINING: PAIRS THAT MEAN THE SAME

a question ↔ its answer

a title ↔ its article

“card was blocked” ↔ “they froze my card”

Pull each pair close, push the rest apart. Over a billion pairs.

“charged twice”

“billed two times”

no word in common

TF-IDF

0

ENCODER

0.80

Closeness is the cosine: **do two vectors point the same way?**

`03-clustering-texts.ipynb`

Demo 03

What are ten thousand complaints about? Real texts sent to the US regulator.

THE RULE OF THE DEMO

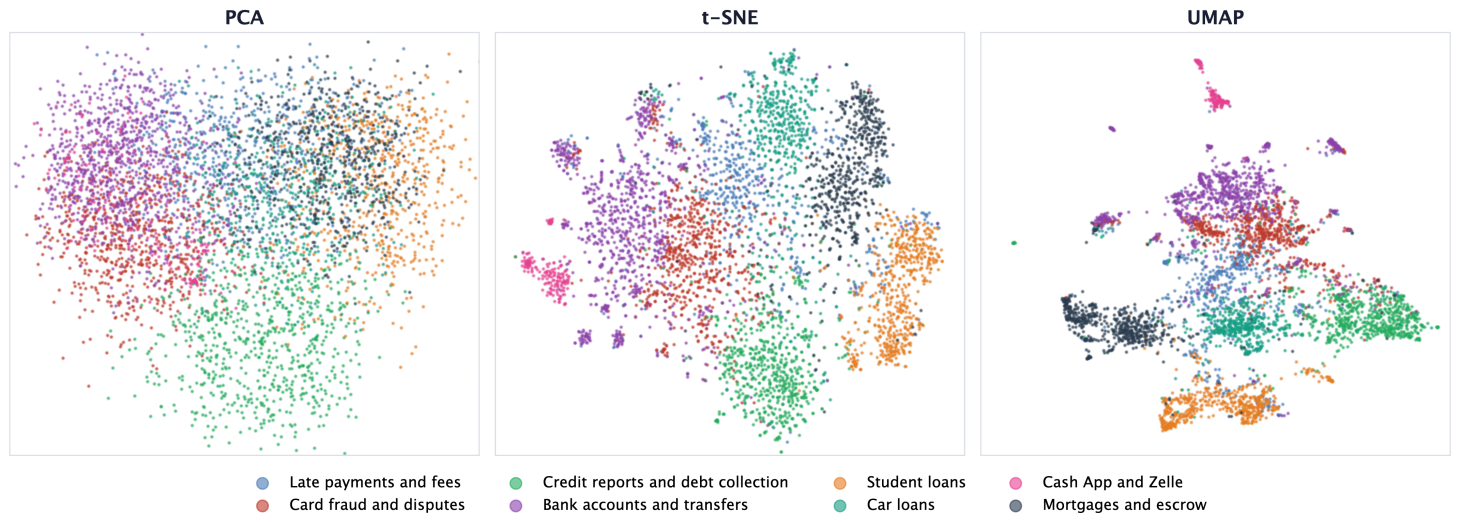
A cluster is real once you can name it.

Read the texts, then name.

1	Complaints, product column hidden
2	Toy texts: TF-IDF vs encoder
3	Encode, find neighbours and twins
4	k-means: **read and name** the clusters
5	Compare with the official products
6	Three maps: PCA, t-SNE, UMAP
7	The January spike: **trend or one month?**

Ask Which product grew most this year? Look at the line first.

A map is not evidence.



Island sizes mean nothing.

Gaps between islands may mean nothing.

Settings change the picture.

The proof is a table, not the map.

distill.pub/2016/misread-tsne pair-code.github.io/understanding-umap

Not “pick an algorithm and go”. Ten steps.

01 **Question and action.** No action, no clustering.

02 **The unit.** A client, a day, a share, a text?

03 **Numbers.** Standardize; texts: TF-IDF or an encoder.

04 **Too many columns?** PCA, or UMAP for texts.

05 **Algorithm by shape.** Round: k-means. Unknown: HDBSCAN.

06 **Settings.** Elbow and silhouette, then: can I name them?

07 **Read the clusters.** Centers, typical words, closest examples.

08 **Stability.** Another seed, another sample.

09 **Name, then check.** An assistant names, you read.

10 **Report.** A table and one picture; the conclusion in words.

The full checklist is card 5 in `concept-cards.html`.

complaints-report.html ↗

Week one, I promised.

Every number in the report was computed in this course. The assistant wrote the HTML around them. We checked what went in.

SESSIONS 1 TO 3
Counts and charts

TONIGHT
The spike we refused to call a trend

TONIGHT
Themes and the map

Conclusion, number, action. **Your HW2 memo works the same way.**

What you can now do.

07	**Clustering and reports**
06	← **Evaluating models**
05	← **Building models**
04	← **Testing ideas**
03	← **Honest charts**
02	← **Joining tables**
01	← **Loading data** ← **understanding the process**

Underneath: **think → ask well → check.**